

◆ 16.8 Control and Feedback Execution Layer

16.8.1 Action Operator

$$B : \mathcal{U}_{\{\text{feasible}\}} \rightarrow \mathbb{R}^d, \\ \quad \|B(u)\| \leq B_{\max}$$

where:

- $B(u)$ maps a decision u to a physical control action
- B is bounded and Lipschitz continuous

Thus:

control actions are well-defined and physically realizable.

16.8.2 Control Injection

$$F \mapsto F + B(u_k)$$

Thus:

decisions directly modify system dynamics.

16.8.3 Full Closed-Loop Structure

Ψ_k
 $\rightarrow P_k$
 $\rightarrow PAC_k$
 $\rightarrow L_k$
 $\rightarrow D_k$
 $\rightarrow u_k$
 $\rightarrow \Psi_{k+1}$

Thus:

the system operates as a fully closed-loop architecture.

16.8.4 No Open-Loop Condition

$u_k = \pi(\Psi_k), \quad$
 $\Psi_{k+1} = F(\Psi_k, \pi(\Psi_k))$

Thus:

the system is self-contained with no external control dependency.

16.8.5 Closed-Loop Stability Preservation

$$\frac{d\mathcal{L}}{dt} \leq -c \|\nabla \mathcal{E}\|^2 + C\sigma^2$$

Thus:

control actions do not violate Lyapunov stability.

16.8.6 Input-to-State Stability (ISS)

$$\|\Psi_k\| \leq \beta(\|\Psi_0\|, k) + \gamma(\|u\|)$$

Thus:

the system remains bounded under control inputs.

16.8.7 Noise Robustness

$$\mathbb{E}[\|\Psi_k\|^2] \leq C + O(\sigma^2)$$

Thus:

stability is preserved under stochastic disturbances.

16.8.8 State Constraints (Saturation)

$$\Psi \in [\Psi_{\min}, \Psi_{\max}]$$

Thus:

physical constraints are enforced.

16.8.9 Fail-Safe Mechanism

$$\Psi_k > \Psi_{\text{crit}}$$

; $\rightarrow$

$$\Psi_k \rightarrow \Psi_{\text{safe}}$$

Thus:

the system avoids catastrophic failure.

16.8.10 Deterministic Limit

$\sigma \rightarrow 0$
;
deterministic dynamics

16.8.11 Real-Time Consistency

Ψ_{k+1} depends only on Ψ_k

Thus:

no delay mismatch occurs in system evolution.

16.8.12 Computational Feasibility

Complexity = $O(N + E)$

Thus:

the system is scalable and real-time implementable.

16.8.13 Integration with System Components

- From Section 16.7: control input u_k
- From Section 16.2: control enters dynamics via F
- From Section 16.3: stability preserved through Lyapunov structure

Thus:

the control layer is fully integrated within the system.

Closure Statement (Part VIII)

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\boxed{
\text{The system is fully closed-loop, stable, and
physically executable.}
}
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SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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